SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	**Assessment:**	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	**Total Marks:**	100
CO5 Statement:	*Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.*		
Student Name:	S.Roshni	**Reg. No.:**	192525185
Section / Batch:	AI&ML	**Date:**	31/08/2026

Code of Conduct

I S.Roshni____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.Roshni______________

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
		GRAND TOTAL:	**100 Marks**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ______________	Signature: ______________	Signature: ______________
Date: _________________	Date: _________________	Date: _________________

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QUESTIONS

Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

1/5 answered

Q1 Minimum Spanning Tree - Kruskal's Algorithm

20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

a. All zones must be connected

b. Total installation cost must be minimized

c. No redundant connections (no cycles) allowed

d. The network must remain fully connected

Your Answer

```
START
Input: Graph with vertices and weighted edges
1. sort all edges in increasing order of weight.
2. create an empty MST.
3. For each edge (u, v):
        If adding the edge does NOT form a cycle:
```

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Q2
Graph - Topological Sort
20 marks

Q3
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20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

2/5 answered

Q2 **Graph - Topological Sort** 20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.
Constraints:
a. Detect cycle before sorting
b. Use DFS approach

Your Answer

```
START
1. Mark all vertices as NOT VISITED.
2. For each vertex:
        If it is not visited:
            perform DFS (vertex).
3. In DFS (vertex):
```

Save Answer

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Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

Q3 **Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's** 20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.

Constraints:

a. Use MST for infrastructure

b. Use shortest path for routing

Your Answer

```
    Update distances of its neighboring nodes.
    Mark the node as visited.
6. Repeat until all nodes are visited.
7. Display:
     MST and its total cost.
     Shortest path and minimum distance.
END
```

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Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

4/5 answered

Q4 **Shortest Path Algorithm - Dijkstra's Algorithm** 20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

```
START
1. set distance of s = 0.
   set all other distances = INFINITY.
2.create two distances for each node:
     dist[node][0] = voucher not used
     dist[node][1] = voucher used
```

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20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q5 **Minimum Spanning Tree - Prim's or Kruskal's Algorithm** 20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST.
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

```
START
1. Find the MST using Kruskal's Algorithm.
2. Set secondBestCost = INFINITY.
3. For each edge in the MST:
        a. Remove that edge from the MST.
        b. Find the next minimum-cost edge
```

Save Answer ✓ Saved